

DEUTSCHE BANK INDIA

Retail Banking — Digital Payments

ACTIVITY DIAGRAMS

UPI Enablement Platform

UML Activity Diagrams — Detailed Control Flow with Parallel Processing

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Table of Contents

Table of Contents2

1. Introduction3

1.1 Notation Key3

2. Send Money — Parallel Validation & Submission4

3. PIN Entry & Lockout Handling5

4. Collect Request Lifecycle6

5. QR Payment Validation.....7

6. Notification Dispatch with Fallback.....8

7. Dispute Investigation & Escalation.....9

1. Introduction

This document presents UML Activity Diagrams for the six most algorithmically complex workflows in the Deutsche Bank India UPI Enablement Platform — flows involving parallel processing, retry loops, or multi-path decision logic that benefit from a level of procedural detail beyond the BPMN process flows and sequence diagrams. Activity diagrams use fork/join bars to show concurrent processing (e.g., balance validation and fraud scoring executing simultaneously) and decision diamonds to show branching logic, including loops back to earlier steps (e.g., PIN retry).

1.1 Notation Key

Symbol	Meaning
Filled Circle	Initial Node — the starting point of the activity
Circle with Ring	Final Node — a terminal outcome of the activity
Rounded Rectangle	Action — a discrete step performed by the actor/system indicated by its colour
Diamond	Decision Node — a branch point with labelled outgoing paths
Solid Black Bar	Fork / Join — a fork bar splits flow into concurrent branches; a join bar synchronises them before continuing
Dashed Arrow	Alternate, exception, retry, or conditional flow

Each diagram is traceable to its corresponding Use Case(s), Functional Requirement IDs (FR-xx), and Business Rules (BR-xx) defined in the FRS.

2. Send Money — Parallel Validation & Submission

This activity models the core payment validation logic, showing balance validation (CBS/CIF) and fraud/velocity risk scoring (Fraud Monitoring) executing in parallel via a fork/join pair before the transaction is either submitted to NPCI or declined. It also models the timeout branch, where a deferred status check determines whether the transaction resolves to Success or requires an automatic reversal.

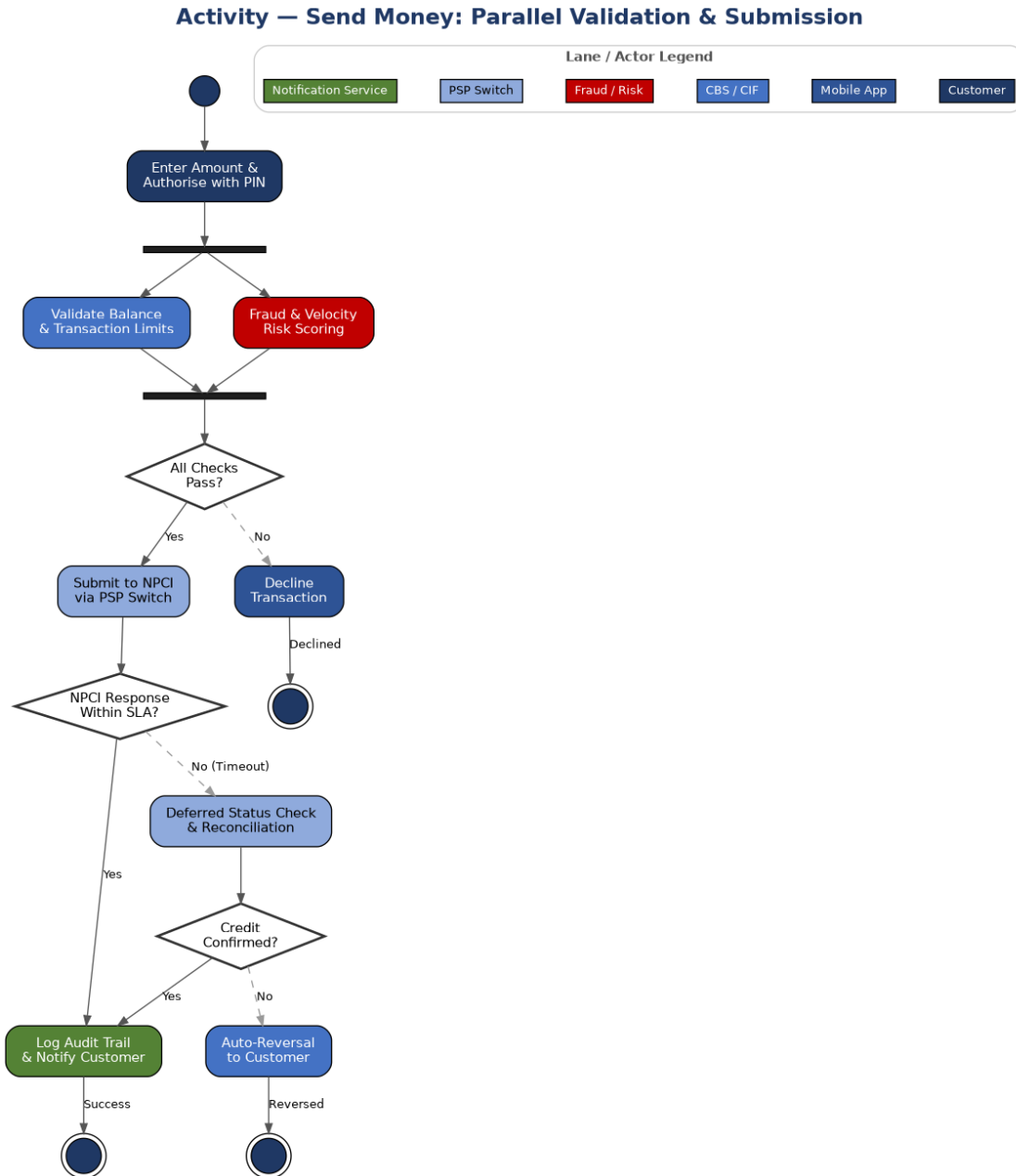


Figure 1 — Activity: Send Money, Parallel Validation & Submission

Related Use Cases / Requirements: UC-10 to UC-14; FR-TXN-07 to FR-TXN-11, FR-TXN-14; BR-TXN-05, BR-TXN-06, BR-TXN-08

3. PIN Entry & Lockout Handling

This activity models the incorrect-PIN retry loop shared by every PIN-authenticated action in the platform: each incorrect entry increments a failed-attempt counter, with a loop back to retry until the third consecutive failure triggers a 24-hour lockout and a mandatory security notification.

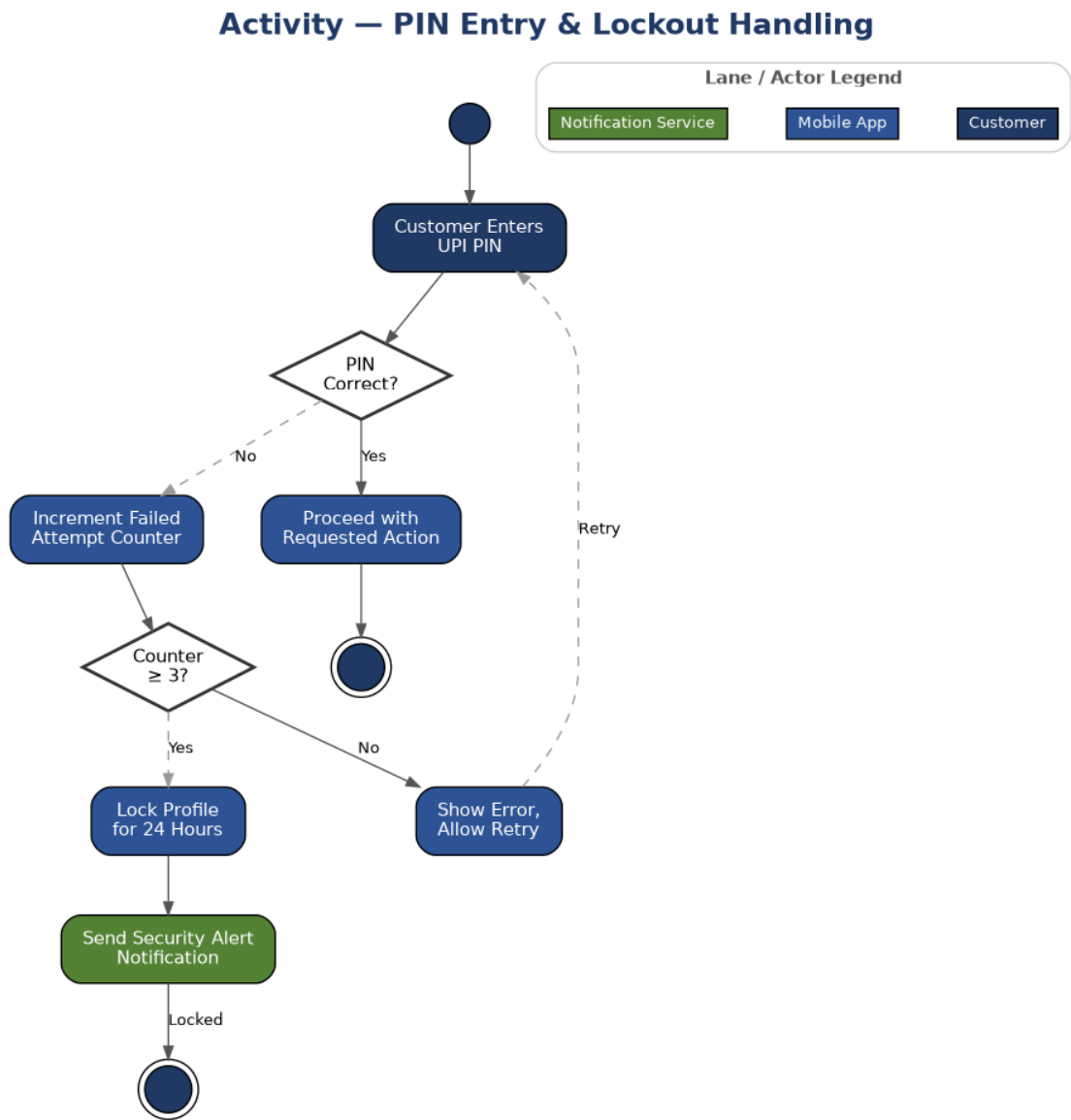


Figure 2 — Activity: PIN Entry & Lockout Handling

Related Use Cases / Requirements: UC-09; FR-PIN-07, FR-PIN-08; BR-PIN-03

4. Collect Request Lifecycle

This activity models the full lifecycle of a collect request from creation through its three possible terminal outcomes (Approved, Rejected, Expired), including the reminder loop available to the requester while the request remains pending.

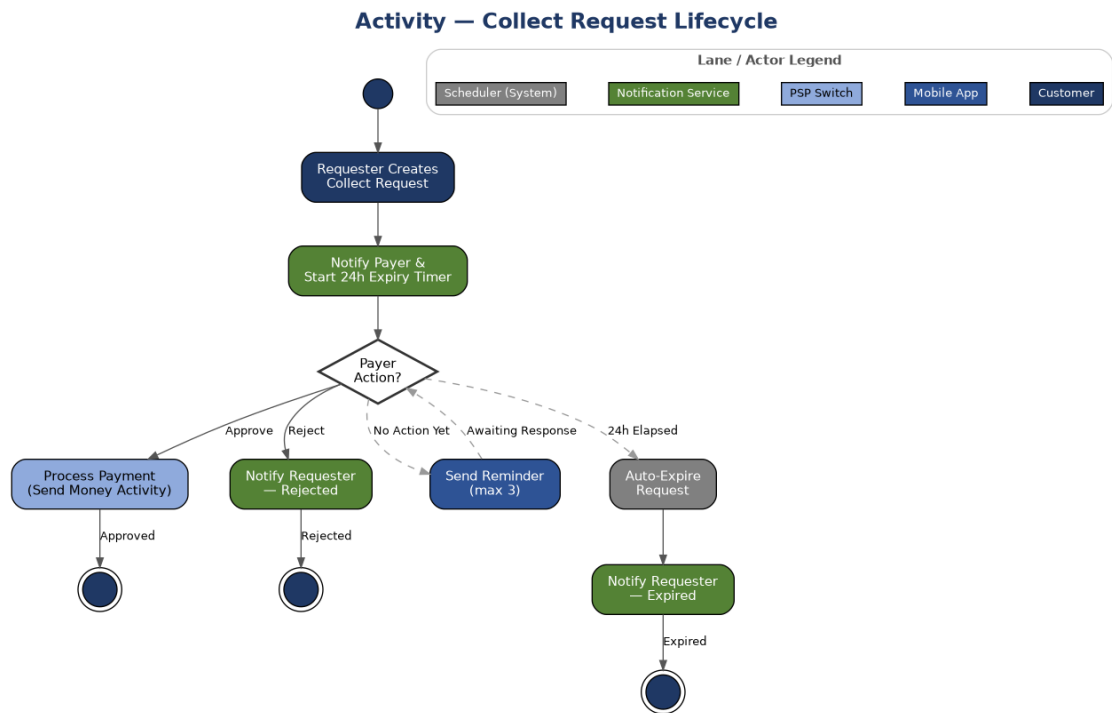


Figure 3 — Activity: Collect Request Lifecycle

Related Use Cases / Requirements: UC-15 to UC-18; FR-COL-01 to FR-COL-08; BR-COL-01, BR-COL-04

5. QR Payment Validation

This activity models QR-type branching (Static/Dynamic P2P vs. Merchant) and, for merchant QR codes, a fork/join pair running NPCI merchant registry validation and internal blacklist checks in parallel before allowing the payment to proceed to the standard Send Money activity.

Activity — QR Payment Validation

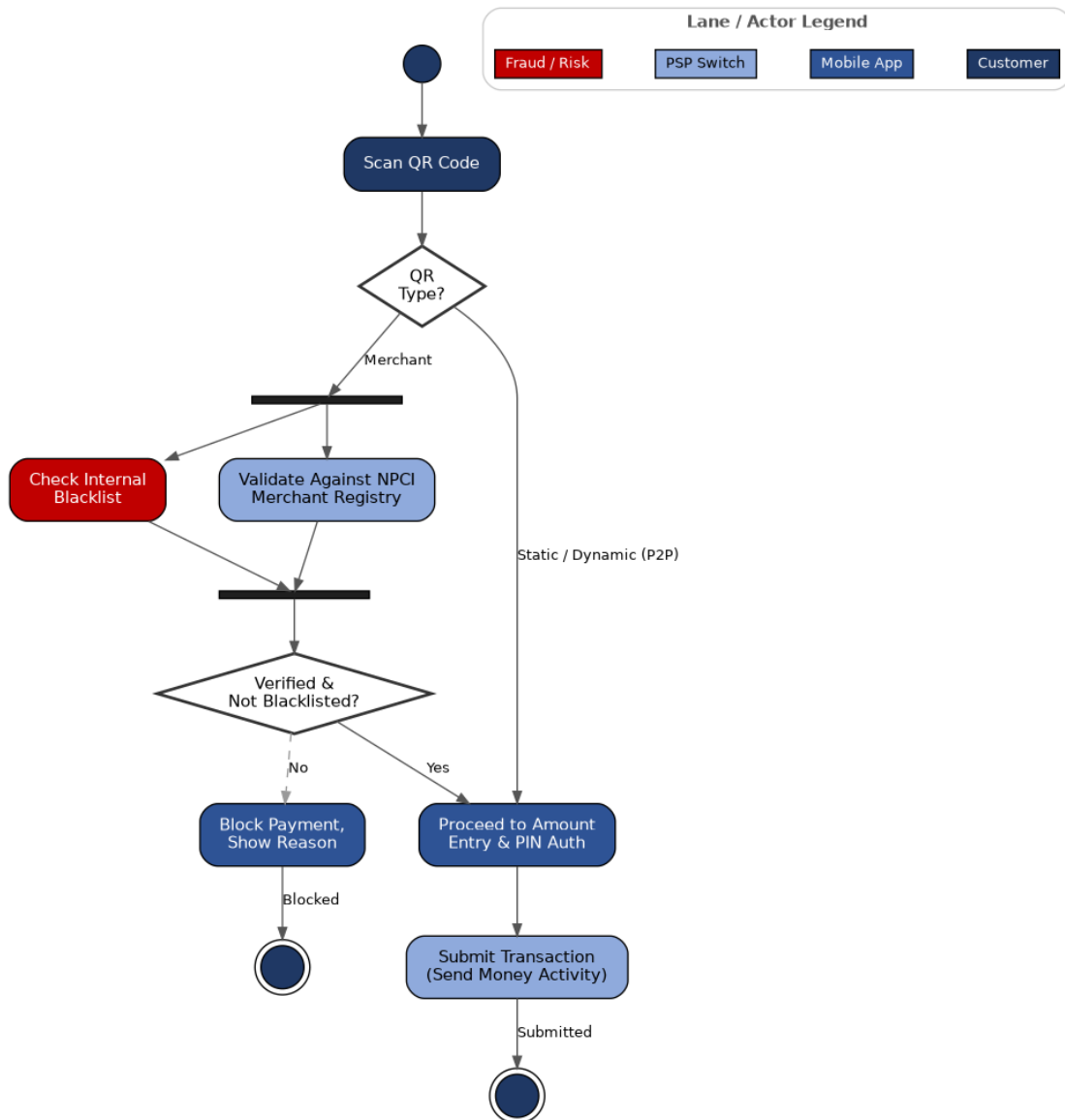


Figure 4 — Activity: QR Payment Validation

Related Use Cases / Requirements: UC-19, UC-20; FR-QR-03, FR-QR-04, FR-QR-06; BR-QR-03, BR-QR-04

6. Notification Dispatch with Fallback

This activity models the notification delivery logic shared across all event types: an initial push delivery attempt, with a conditional SMS fallback specifically for security-critical events if push delivery fails, ensuring critical alerts are never silently dropped.

Activity — Notification Dispatch with Fallback

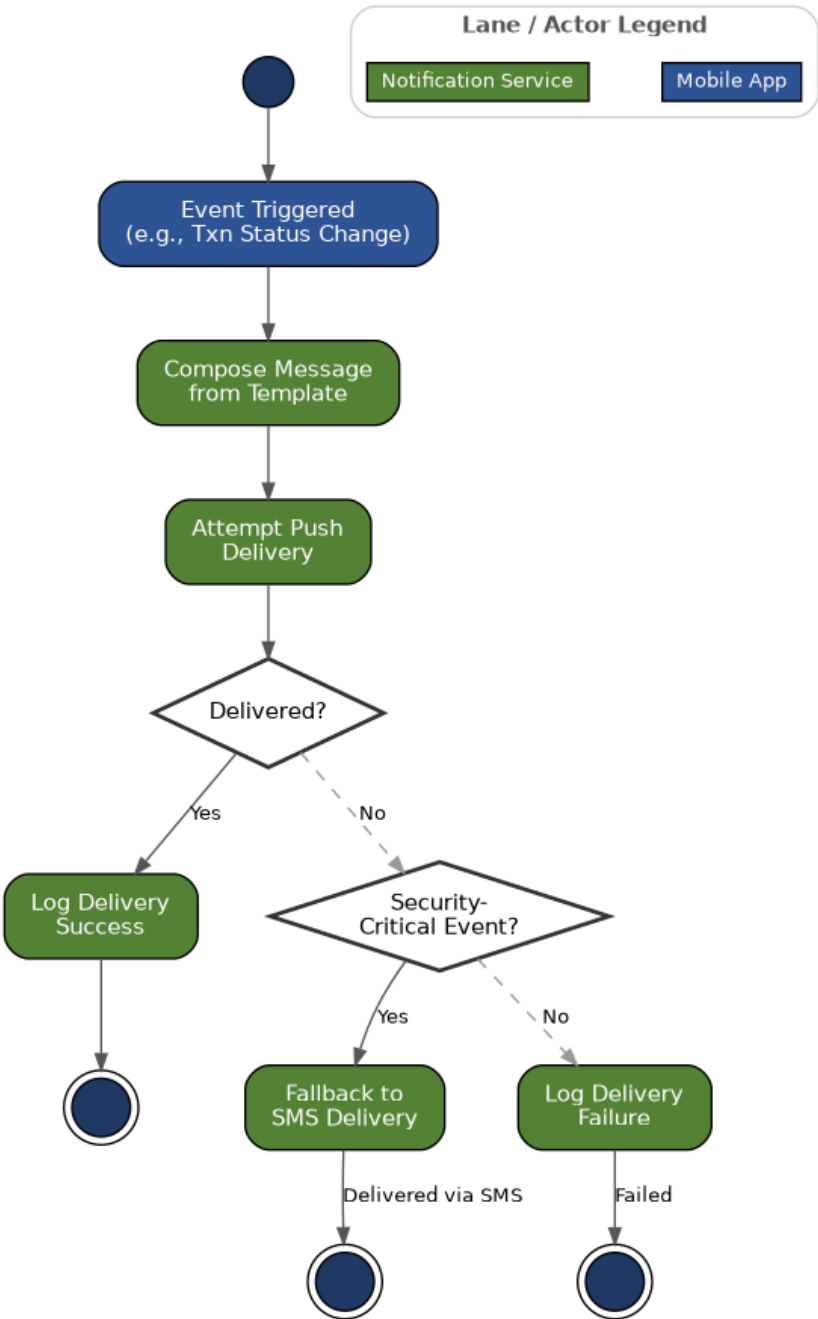


Figure 5 — Activity: Notification Dispatch with Fallback

Related Use Cases / Requirements: UC-24; FR-NOT-01 to FR-NOT-06; BR-NOT-01

7. Dispute Investigation & Escalation

This activity models conditional NPCI escalation (fork/join with internal investigation running in parallel where applicable), the scheduled TAT-breach check that can auto-escalate a stalled investigation, and the final refund/reject decision leading to customer notification.

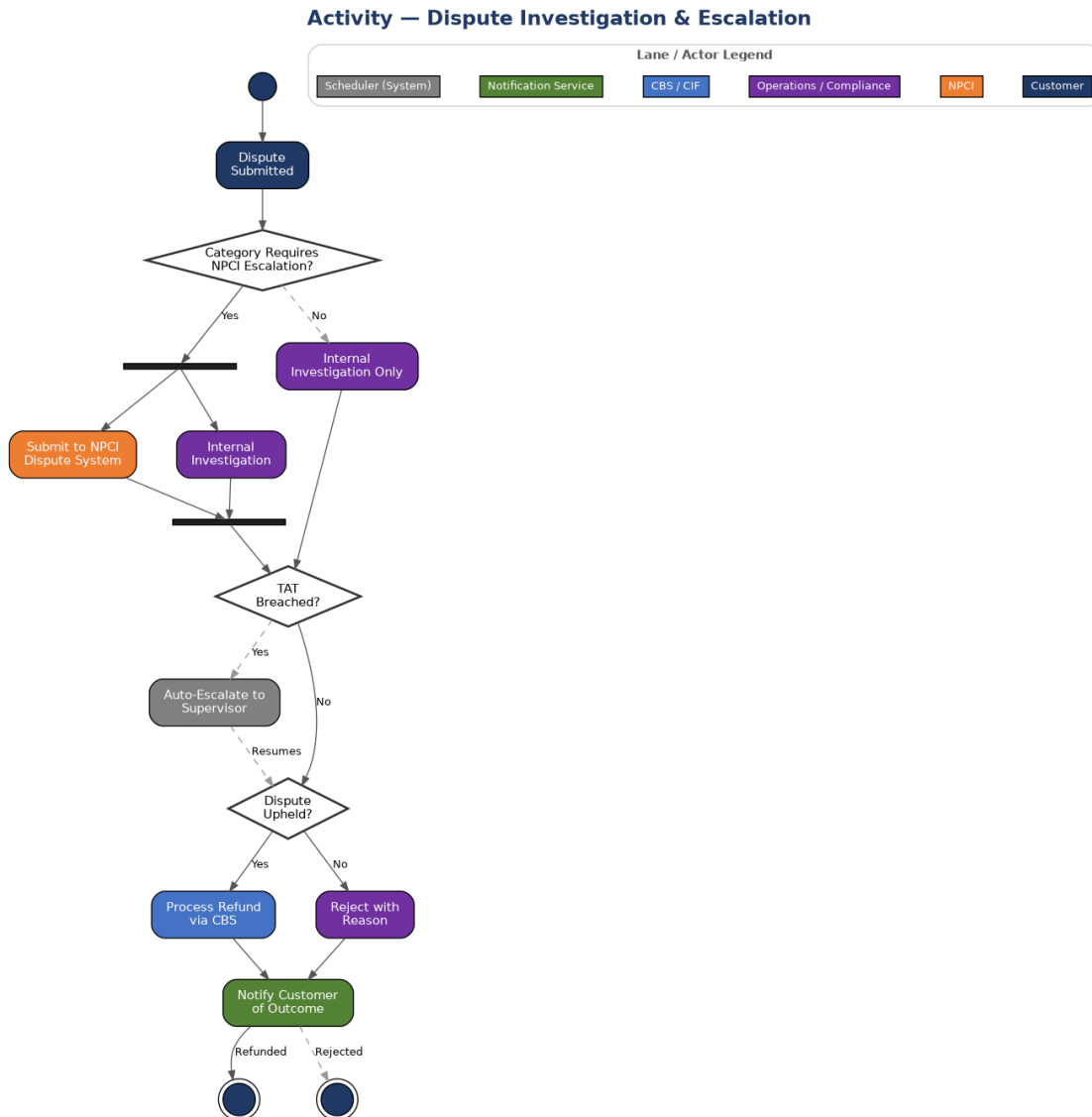


Figure 6 — Activity: Dispute Investigation & Escalation

Related Use Cases / Requirements: UC-26, UC-28, UC-29; FR-DIS-01 to FR-DIS-08; BR-DIS-01, BR-DIS-03